Jorge Nava de la Peña

Senior Software Engineer · Applied AI (LLM agents, tool use, evals)

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SUMMARY

Senior software engineer with 8+ years shipping production systems applying AI since 2021 (IBM Watson) and building LLM applications since 2025: multi-agent orchestration, tool use with human confirmation, MCP integrations and evaluation harnesses. Currently at Wizeline, embedded with Realtor.com on a platform that syndicates 1.6M+ listings. Founder of One Spark, a consultancy where I own client work end to end, from discovery and pricing to architecture and delivery. Bilingual (English/Spanish).

EXPERIENCE

Wizeline — Software Engineer → Senior Software Engineer (promoted 2026) Jul 2023 – Present · Guadalajara

Realtor.com (Move Inc. / News Corp) · ListHub Syndication · Jul 2025 – Present

- Own AppSec remediation across 10+ microservices that distribute 1.6M+ listings from 575+ data sources to 120+ publishers for a site with ~60M monthly unique visitors.
- Closed 5 AppSec tickets in a single PR by modernizing the ListHub email service (Hibernate 3.6→5.6, Jetty 7.6→9.4, log4j 1.x→2.x, Servlet 2.5→3.1) and removing XSS and SQL-injection paths.
- Traced 6 syndication cronjobs stretching from 3 h to 10–22 h (~2,304 GiB-hrs/day of cluster pressure) to a request-equals-limit CPU/memory anti-pattern and led the rightsizing initiative.
- Found 10+ production S3 buckets serving unencrypted HTTP (one handling 39.7M requests) and moved secrets from Vault to AWS Secrets Manager with cross-account IAM trust.
- Merged ~350 pull requests across 50+ repositories in 16 months, working daily with AI-assisted tooling (Claude Code, Gemini, Glean).

Valvoline & Boats Group · Jul 2023 – Jun 2025

- Led the Shoes For Crews pilot release for Valvoline as the main point of contact with the client: gathered requirements, designed the solution, managed a small dev team and coordinated go-live with client and third-party teams.
- Moved legacy endpoints to Spring Boot services and upgraded services from Spring Boot 2.x to 3.x; unblocked a cross-system CarFax API integration through pair programming and QA validation.
- Reviewed 30+ pull requests per month, with more than 10% proposing refactors or alternative architectures.
- Boats Group: built multi-tenant React (SSR) + Node.js features on AWS (Terraform, Docker, GitHub Actions) for display ads, SEO redirects and customer reporting.

One Spark — Founder, CEO & Principal Engineer (part-time, alongside full-time roles) Jan 2018 – Present · Zapopan

- Run a digital consultancy for 10+ clients in retail, healthcare, hospitality and fintech: discovery, pricing, architecture, delivery and support, leading a team of up to 4 developers.
- **SAM, AI content platform (product):** built a multi-tenant system where an assistant plans and drafts brand content through tool calls, with a two-phase confirmation before any action executes. Added best-of-N generation with LLM judges for text and images and an eval harness (deterministic checks + LLM-judged golden brand profiles); captions scored 8.5/10 on two real brand profiles. 267 automated tests.
- Long-term technology partner of a multi-branch jewelry retailer: Shopify stores for Mexico and the US plus a serverless AWS backend (Lambda, API Gateway) of 10 Lambda functions for inventory sync, price monitoring, appointments and repairs, running ~3,800 automated jobs a month for a store with ~25k weekly sessions.

IBM — Full Stack Developer → Technical Owner → Jr. Solution Architect Jul 2021 – May 2023 · Guadalajara

Software Developer Intern · Feb 2021 – Jul 2021

- Automated purchase-order and financial-report creation for internal finance teams using IBM Watson and multi-system orchestration, and built tools for faster financial reconciliation.
- As Technical Owner, aligned operational processes with compliance and standardization requirements.
- Designed CI/CD pipelines on IBM Cloud and Red Hat OpenShift and maintained cloud infrastructure (load balancing, LogDNA monitoring).

PrimalTechnologies — Software Developer & Solution Architect Jun 2020 – Feb 2021 · Remote (Ciudad Juárez office)

- Built Node.js APIs and Dockerized cloud infrastructure on DigitalOcean with SQL and NoSQL databases.

Rocket Publicidad — Full Stack Developer Jan 2020 – Jun 2020 · Remote

APPLIED AI PROJECTS (PRODUCTION, PERSONAL)

- **Nava OS assistant:** replaced an intent classifier and 7 hand-written handlers with a single tool-use orchestrator on AWS Lambda, answering query types the old design could not; integrates calendar, GitHub and notes with confirmation before writes.
- **Tax & finance agents:** two cooperating agents with shared memory and a coordination channel, working in copilot mode; a verifier sub-agent checks every tax rule against the literal text of official sources and flags anything unverified instead of guessing.
- **Docs RAG with evals (open source, in progress — Dec 2026):** retrieval-augmented Q&A over public AWS documentation with hybrid search (BM25 + embeddings), cross-encoder reranking and cited answers, measured by a curated 100-question eval set (retrieval recall@5 and LLM-judged faithfulness) gated in CI.

SKILLS

Applied AI: Claude API (tool use, prompt caching), multi-agent orchestration, MCP, evals / LLM-as-judge, best-of-N, prompt engineering

Languages: TypeScript, Python, Java, JavaScript, SQL, Go

Cloud & infra: AWS (Lambda, S3, EC2, RDS, DynamoDB, CloudFront, SAM, CDK, IAM, Secrets Manager), EKS, Helm, ArgoCD, Docker, CircleCI, Vercel, OpenShift

Web & data: React, Next.js, Node.js, Spring Boot, PostgreSQL, Supabase, DynamoDB, Redis, OpenSearch, Snowflake, Shopify

EDUCATION & CERTIFICATIONS

B.S. Software Engineering, Universidad Autónoma de Guadalajara, 2022 — Honors.

AWS Certified Solutions Architect – Associate (2024) · IBM Containers & Kubernetes Essentials (2022) · IBM Watson foundations (2021)